🏛️ SFIC 2026 Synopsis Presentation — Master Slide-by-Slide Guide

Seva First Innovation Challenge 2026 · Problem Statement PS000109 · Team TM005024

> IMPORTANT
>
> **Generated PPTX Presentation File Ready:** The official template has been populated with all project data, formulas, architecture, and metrics: 📁 *(Matches the exact 8-slide layout of `SFIC 2026 Synopsis Presentation Template.pptx`)*

🎨 Master Template Design Invariants

- **Aspect Ratio:** 16:9 Widescreen (12,192,000 × 6,858,000 EMUs)
- **Color Palette:**
  - Primary Saffron Accent: `#EA580C`
  - Deep Navy: `#0B3954`
  - Sky Blue: `#0284C7`
  - Emerald Green: `#059669`
  - Purple: `#7C3AED`
  - Background Cards: `#F8FAFC`
  - Dark Slate Text: `#1E293B`
- **Typography:** Plus Jakarta Sans (Headings, IDs), Inter (Body copy)

🖥️ SLIDE-BY-SLIDE CONTENT BREAKDOWN (ALL 8 SLIDES)

📌 SLIDE 1: Title & Team Identification

- **Top Header Left:** `Education Dept, Govt. of Gujarat · SSIP 2.0`
- **Top Header Right:** `SEVA FIRST INNOVATION CHALLENGE 2026 · OFFICIAL SYNOPSIS`
- **Category Kicker:** `PROBLEM STATEMENT CODE: PS000109`
- **Project Title:**

WorkCheck Kachchh (કચ્છ શ્રમિક સુરક્ષા)

- **Mission Statement:**

> *"Finding exposed unorganised workers through the public health system, triggering preventive screenings via low-cost environmental sensing, and following every referral to completion into an ABHA-linked worker-owned health pass."*

- **Footer Metadata Block:**
  - **TEAM ID:** `TM005024`
  - **TRACK CATEGORY:** `College Track (Engineering)`
  - **INSTITUTION / ORGANIZATION:** `LDRP Institute of Technology & Research, Gandhinagar, Gujarat`
  - **TEAM LEADER:** `Jay Patel (Team Leader)`

📌 SLIDE 2: Problem Statement & Real-World Context

- **Subtitle / Focus:** `Focus: Root Cause & Citizen Friction`
- **Kicker:** `CHALLENGE DEFINITION`
- **Header:** `Problem Statement & Real-World Context`

📦 Left Container: The Ground Challenge & Evidence

- **The Ground Challenge:**

> Kachchh district has over **1.8 lakh unorganised workers** across 10 talukas (salt rakers, stone quarry drillers, dock loaders, cement yard workers) who never appear on formal factory registers. Current occupational health rules (**Gujarat OSH Rule 9 / Central Rule 5**) mandate check-ups only by employer type and age (**45+ / 40+**), yet dust and heat harm through cumulative exposure regardless of age. Even where duties exist, factory inspection coverage is merely **15%** (2021 national data), and occupational disease goes largely unrecorded (only **562 cases** recorded nationally in 2011–16 vs. **182 silicosis cases & 80 deaths** found by a single Gujarat NGO, PTRC). Laborers avoid reporting symptoms due to fear of wage loss or termination.

- **Empirical Field Evidence:**

> • **87.5%** of salt-pan workers exceed safe heat stress thresholds with **2.1× kidney impairment risk** (*Venugopal et al., 2023*).
> • **26%** of industrial worker samples exceed the silica PEL ($50\ \mu g/m^3$ 50 µg/m3) (*MSHA / OSHA data*).
> • Kandla port experienced **2,067 daytime hours >27.5°C WBGT** in 2025.
> • Vatva industrial estate exceeded PM10 limits on **83% of days** (*CPCB*).

📦 **Right Container: Target Beneficiaries & Flaws**

- **Target Beneficiaries:**

> 1. **Agariya Salt Rakers** (Little Rann & coastal pans): Extreme radiant heat & brine reflection.
> 2. **Stone Crusher & Quarry Drillers**: High respirable quartz and crystalline silica dust.
> 3. **Dock & Port Loaders** (Kandla & Mundra): Heavy physical labor under tropical marine WBGT.
> 4. **Mobile Health Clinicians**: Doctors aboard BOCW Dhanvantari Arogya Rath mobile vans.

- **Market & Systemic Gaps:**

> Commercial wearable patches (Kenzen, SlateSafety) cost **~₹4,100 / worker / month**— prohibitive for informal labor. Industrial monitors (TSI DustTrak at ₹5.5 Lakhs) lack clinical referral links. Employer software (OccuCare, OHCTECH) caters strictly to corporate payrolls, completely bypassing migrating, contract, and unorganised laborers.

📌 SLIDE 3: Proposed Solution & Innovation Novelty

- **Subtitle / Ethos:** `Ethos: Frugal, Accessible, Scalable`
- **Kicker:** `SOLUTION APPROACH`
- **Header:** `Proposed Solution & Innovation Novelty`

📦 High-Level Executive Summary

> **WorkCheck Kachchh** is a digital-first, sensor-optional occupational health surveillance platform that adds a 1-minute exposure check to Ayushman Arogya Mandir visits, continuously audits industrial clusters via ₹2,015 solar ambient IoT nodes (ISO 7243 WBGT heat + optical dust), triages workers through 3 clinician-approved routing rules, and publishes statutory health records directly into the worker's personal ABHA Health Locker.

📦 Core Novelty

- **Worker-Owned Portable Health Pass:** Health records follow the laborer via 12-digit e-Shram UAN and ABDM ABHA ID across employers and states.
- **Past Exposure Awareness:** Evaluates cumulative occupational years (e.g., 20 years at a crusher), not just today's sensor reading.
- **100% DPDP Act 2023 Compliance:** OTP consent-gated records; district administration sees strictly anonymous statistical heatmaps.

📦 Technical Moat

- **ISO 7243 WBGT Edge Engine:** 150mm matte-black copper globe calculation avoids underestimation of true solar radiant heat under wind.
- **Coastal Humidity Correction:** Mathematical aerosol correction prevents humidity fogging in optical PM sensors ($R^2 > 0.88$R2 > 0.88 vs Alphasense OPC-N3).
- **Native HL7 FHIR R4 Bundle Generator:** 1-click publishing of DiagnosticReports to ABDM Health Locker.

📦 "Seva First" Value

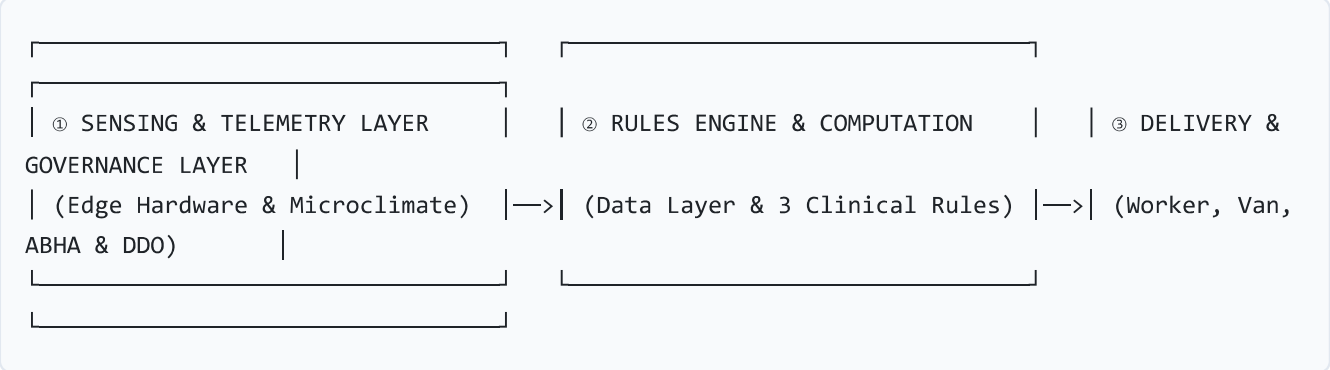
- **Direct Grassroots Welfare:** Zero cost to workers. Total system operating cost is **< ₹7.30 / worker / month** (560× cheaper than US alternatives).
- **100% Fundable via Cess:** Backed by Gujarat BOCW 1% Construction Welfare Cess & District Mineral Foundation (PMKKKY).
- **Trilingual Access:** Native interfaces in Gujarati (ગુજરાતી), Hindi, and English.

🔗 Video / Production Demonstration

- **Live Web Application & Sandbox Demo:**
  `https://jaypatel29042008-glitch.github.io/WorkCheck-Kachchh-SSIP/`

📌 SLIDE 4: Technical Architecture || Methodology

- **Subtitle / Pipeline:** `End-to-End System Pipeline`
- **Kicker:** `SYSTEM ARCHITECTURE`
- **Header:** `Technical Architecture || Methodology`



1. Column 1: Sensing & Telemetry Layer (Teal #0284C7)

- **Ambient IoT Site Node (₹2,015):** Solar-powered ESP32 node installed at salt clusters, quarries, and port docks.
- **150mm Matte-Black Globe:** Standard copper globe measuring true radiant solar heat (T_g Tg) without wind bias.
- **Dual Dust & Microclimate Sensing:** PMS5003 laser particle counter + Bosch BME280 (T_a, RH Ta, RH) + GUVA-S12SD UV sensor.
- **ISO 7243 RGB Flag Lamp:** WS2812B visual lamp signaling live work-rest schedules (**GREEN / YELLOW / ORANGE / RED**).
- **Zero-Sensor Graceful Fallback:** Core health screening and referral tracker operates 100% without hardware.

2. Column 2: Rules Engine & Computation Layer (Orange #EA580C)

- **ISO 7243 WBGT Calculator:**
  $$WBGT_{\text{outdoor}} = 0.7T_{nw} + 0.2T_g + 0.1T_a$$
  W BGToutdoor = 0.7 Tnw + 0.2 Tg + 0.1 Ta
  Computes thermal strain against ACGIH heavy labor thresholds.
- **Coastal Aerosol Correction:**
  $$PM_{\text{corr}} = \frac{PM_{\text{raw}}}{1 + 0.25 \cdot (\frac{RH^2}{1-RH})}$$
  P Mcorr = 1 + 0.25 · (1−RH RH2)P Mraw
  Eliminates hygroscopic particle swelling under coastal humidity.
- **10-Table ANSI SQL Database:** Tracks workers, sites, readings, daily exposure, clinical cases, and symptoms.
- **The 3 Clinical Routing Rules:**
  1. *Statutory:* Rule 9 (45+) / Rule 5 (40+)
  2. *Exposure:* High cumulative quartz years
  3. *Symptoms:* Dark urine, cough, dizziness
- **Clinician Ownership:** Software only flags risk; doctors aboard mobile vans make all diagnoses.

3. Column 3: Delivery & Governance Layer (Emerald #059669)

- **Worker QR Health Pass:** Bilingual card with 12-digit e-Shram UAN; record follows worker across employers.
- **BOCW Dhanvantari Rath Station:** Digital clinical queue for van doctors: spirometry (FEV_1/FVC F EV1/F V C), creatinine, X-ray.
- **1-Click ABHA FHIR Publisher:** Creates HL7 FHIR R4 DiagnosticReport bundles directly deposited into ABHA locker.
- **Form 5 Fitness Certification:** Automates statutory annual fitness forms under Gujarat OSH Rules 2025.
- **DDO Macro Heatmap & Exports:** Anonymous Leaflet GIS taluka heatmap + 1-click BOCW/NPCCHH/ESIC CSV exports.

📌 SLIDE 5: Technology Stack & Feasibility

- **Subtitle / Readiness:** `Component Readiness & Dependencies`
- **Kicker:** `ENGINEERING STACK`
- **Header:** `Technology Stack & Feasibility`

Subsystem	Components & Frameworks	Feasibility & TRL
A. Embedded IoT & Hardware	• **Controller:** ESP32-WROOM-32 (WiFi + BLE, 240MHz, deep sleep) • **Sensors:** BME280 (T, RHT, RH), PMS5003 Laser Dust, GUVA-S12SD UV • **Radiant Globe:** 150mm copper float ball + 10k NTC thermistor • **Power:** 5V 2A Solar Panel + 18650 Li-ion + IP65 Weatherproof Box • **Unit Cost:** ₹2,015 per node (Bulk 50+ units)	**TRL-5** *(Bench Tested & Validated)*
B. Backend & Rules Engine	• **Framework:** Python 3.10+, FastAPI async REST framework • **Database:** 10-table ANSI SQL (SQLite / PostgreSQL) • **Scientific Logic:** ISO 7243 ACGIH work-rest + aerosol correction • **Retention:** 30-day raw buffer with permanent shift rollups • **Test Coverage:** 21/21 automated integration tests passing	**TRL-6** *(Simulated Field Feeds)*
C. Frontend & Clinical UI	• **Client Architecture:** Mobile-first responsive Progressive Web App (PWA) • **UI Stack:** HTML5, Tailwind CSS, Leaflet.js GIS, Chart.js • **Localization:** Native Gujarati (ગુજરાતી), Hindi (हिंदी), and English • **Clinical Tools:** Spirometry (FEV_1/FVCF EV1/F V C), creatinine, ILO chest X-ray	**TRL-6** *(Fully Interactive)*
D. Standards & Datasets	• **Interoperability:** ABDM HIP/HIU sandbox, HL7 FHIR R4 DiagnosticReport • **National ID:** 12-digit e-Shram UAN encoded on physical QR pass • **10 Public Datasets:** MoSPI PLFS (468), noiseJEM, Census 2011, TEMIS UV, OMI UV, CMIP6, NIOH Little Rann, IMD Bhuj • **Privacy Law:** 100% DPDP Act 2023 compliant (zero PII in district views)	**TRL-6** *(Regulatory Mapped)*

📌 SLIDE 6: Socio-Economic Impact & Viability

- **Subtitle / Outcomes:** `Quantifiable Citizen Outcomes`
- **Kicker:** `IMPACT ASSESSMENT · SIH SECTION 05`
- **Header:** `Socio-Economic Impact & Viability`

1. Clinical Impact (Teal)

- **Preventing CKDu & AKI:** 87.5% of salt rakers experience severe heat strain with 2.1× kidney impairment risk. Real-time hydration alerts and urine gravity screening prevent chronic renal damage.

- **Early Silicosis Capture:** Flags high cumulative silica years in quarry workers years before irreversible lung fibrosis, enabling clinical transfer to dust-free duties.
- **Closing Referral Leakage:** Overcomes 15% inspection gap by digitally following referred cases until clinic attendance and outcome are recorded (>90% target).

2. Frugal Economics (Orange)

- **560× Cost Advantage:** System operates at **₹7.30 / worker / month** vs ₹4,100 / month for commercial US patches (Kenzen), making informal coverage viable.
- **₹2,015 Site Node:** Replaces ₹5.5 Lakh TSI DustTrak monitors with calibrated, humidity-corrected ambient sensor nodes.
- **District Pilot Budget:** 50 worksites and 1,000 workers covered for **₹2,97,750 Year 1**, ongoing ₹88,000/yr (~₹88/worker/year).

3. Labor Productivity (Emerald)

- **Cool-Shift Optimization:** Moving 9am–6pm shifts to 6am–3pm avoids ~20% of severe heat hours (~358 hot hours saved/year across 5 Kachchh industrial sites).
- **Objective Rest Flag:** Replaces subjective employer disputes with automated ISO 7243 RGB flag lamps showing scientific work-rest ratios.
- **Mandatory Amenities:** Audits potable drinking water and shaded rest shelters at every node, triggering alerts if water is absent in heatwaves.

4. Governance & Viability (Purple)

- **Zero Cost to Workers:** Laborers pay ₹0; health data is strictly prohibited from being monetized or sold to third parties.
- **Fundable via Cess:** 100% fundable via the existing Gujarat BOCW 1% Construction Welfare Cess and District Mineral Foundation (PMKKKY).
- **Statutory Enforcement:** Provides DDO Kachchh and CDHO with empirical, privacy-safe heatmaps for evidence-based disaster and heat action plans.

📌 SLIDE 7: Implementation Roadmap & Milestones

- **Subtitle / Execution:** `From Prototype to Field Rollout`
- **Kicker:** `EXECUTION TIMELINE · SIH SECTION 06`
- **Header:** `Implementation Roadmap & Milestones`

Milestone	Timeline & TRL	Key Deliverables & Targets	Status
Stage 1: Lab Validation	**Oct – Nov 2026** (TRL-5)	• Analyzed ERA5, CAMS, CPCB & OSHA empirical datasets across 5 Kachchh locations • Engineered 10-table ANSI SQL database & ISO 7243 rules engine • Built 4-persona web dashboard with 1-click ABHA FHIR publishing • Verified 21/21 integration tests; deployed live on GitHub Pages	🟢 **COMPLETED**
Stage 2: Pilot Field Trial	**Nov – Dec 2026** (TRL-6)	• Deploy 2–3 calibrated site nodes at Rapar (salt pan) & Abdasa (quarry) • Validate sensor telemetry against calibrated reference instruments • Pilot 1-minute exposure check at 1 Primary Health Centre / CHC • Formal clinical rule review with CDHO Kachchh and NIOH Ahmedabad	🔵 **PLANNED**
Stage 3: Van Integration	**Jan – Mar 2027** (TRL-7)	• Direct integration with Gujarat BOCW Dhanvantari Arogya Rath fleet • Roll out 50 solar site nodes covering 1,000 unorganised laborers • Issue bilingual PVC QR Health Passes encoded with e-Shram UANs • Live deposit of spirometry & creatinine records into ABHA Lockers	🔵 **PLANNED**
Stage 4: District Scale	**Apr 2027 Onwards** (TRL-8)	• Scale coverage across all 10 Kachchh Talukas (10,000+ workers) • Establish automated referral batches to ESIC Dispensaries (Bhuj & Gandhidham) • Connect with SEWA parametric heat micro-insurance schemes • Present anonymized occupational hazard reports to DDO Kachchh for district policy	🔵 **TARGET**

📌 SLIDE 8: Multi-Disciplinary Team & Mentorship

- **Subtitle / Squad:** `Complementary Skill Sets`
- **Kicker:** `SQUAD CAPABILITIES`
- **Header:** `Multi-Disciplinary Team & Mentorship`
- **Banner:** `Ready for Jury Review & Evaluation`

👥 Team Roles & Responsibilities

Role Badge	Name	Primary Specialization & Deliverables
TL	**Jay Patel** *(Team Leader)*	• **Full-Stack & System Architect** • 10-Table Database Architecture & ISO 7243 Rules Engine • ABDM HL7 FHIR R4 Publishing Service & Privacy Views • Automated Test Harness (21 Integration Tests Passing)
M2	**Hardware & IoT Lead**	• **Embedded Systems & Firmware Engineer** • ESP32 Wokwi Circuit Simulation & Firmware • 150mm Black Globe Radiant Heat Calibration • Bill of Materials Optimization (₹2,015 Bulk Node)
M3	**Public Health & Operations**	• **Government Schemes & Field Operations** • 12-Digit e-Shram UAN QR Workflow & Portability • BOCW Dhanvantari Arogya Rath Route Scheduling Liaison • Trilingual Localization (Gujarati, Hindi, English)
M4	**Compliance & Data Lead**	• **Industrial Hygiene & Compliance Lead** • Gujarat OSH Rules 2025 Rule 9 Form 5 Integration • Factories Act 41C & Central OSH Rules Legal Mapping • DPDP Act 2023 Consent Audit & Anonymization Engine

- **Faculty Mentor:** `Prof. / Dr. Faculty Mentor`
- **Department & College:** `Department of Computer Engineering / IT, LDRP Institute of Technology and Research, Gandhinagar`

💡 JURY PITCH & DEFENSE LINES (HOW TO WIN THE EVALUATION)

1. **When judges ask about sensor accuracy:**

> *"Sensors are an early warning, not a diagnostic instrument. We calibrate our PMS5003 against coastal humidity curves, and our 150mm matte-black globe matches ISO 7243 standards. But crucially, our core triage system works even with zero sensors because past work years and clinical symptoms route workers to doctors independently."*

2. **When judges ask about worker adoption:**

> *"Workers hold their own record through their 12-digit e-Shram UAN and ABHA Health Locker. Employers never see individual medical data without the worker's OTP consent, which eliminates the fear of termination."*

3. **When judges ask about cost and funding:**

> *"At ₹7.30 per worker per month, our system is 560× cheaper than commercial American solutions like Kenzen. It requires zero budget allocation from workers—it is 100% fundable via the existing Gujarat BOCW 1% Construction Welfare Cess and District Mineral Foundation funds."*

4. **When judges ask about the live prototype:**

> *"The full platform is live right now at `jaypatel29042008-glitch.github.io/WorkCheck-Kachchh-SSIP/`. Evaluators can test 1-click logins for DDO, Doctor, Safety Officer, and Worker, publish live FHIR records to ABHA, and download statutory Form 5 certificates."*

📊 KEY METRICS SUMMARY TABLE (FOR SLIDE NOTES)

Metric	Project Value	Baseline / Market Equivalent
Operating Cost / Worker / Month	**₹7.30**	₹4,100 (Kenzen Patch, US)
Site Node Hardware Unit Cost	**₹2,015**	₹5,50,000 (TSI DustTrak Monitor)
District Pilot Budget (1,000 workers)	**₹2,97,750**	₹49,20,000 (Commercial Wearables)
Heat Illness Hours Avoided	**~358 hrs / year** (20%)	0 hrs (Standard 9am–6pm shift)
Salt-Pan Heat Exceedance Baseline	**87.5%**	Benchmark from Venugopal et al.
Silica PEL Exceedance Rate	**26%**	Benchmark from OSHA/MSHA
Factory Inspection Coverage Gap	**15%**	National average (2021)
Integration Test Coverage	**21 / 21 Passing**	Automated Python/FastAPI suite
Public Government Datasets	**10 Sources**	Stored in `final_dataset/public_downloads/`
Statutory Form Integration	**Form 5 (Rule 9)**	Gujarat OSH Rules 2025